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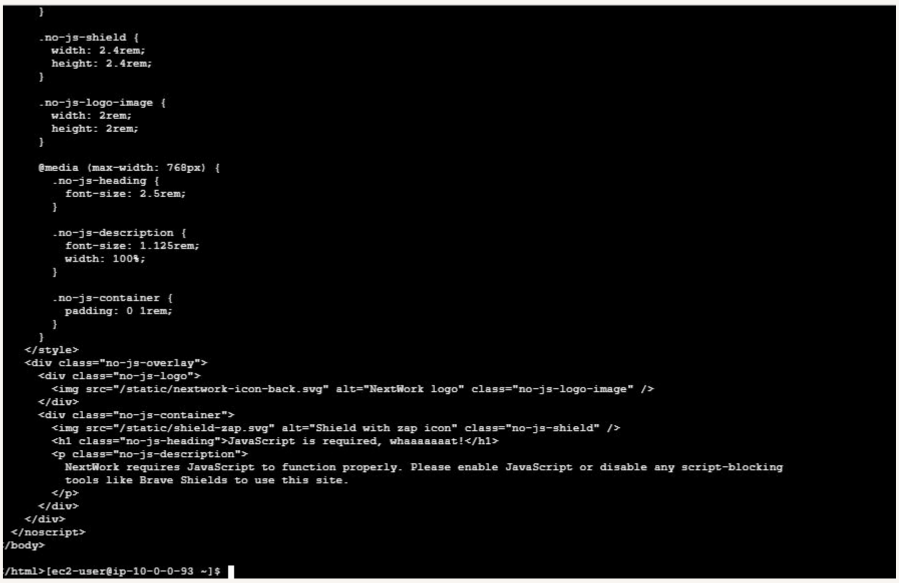
# Testing VPC Connectivity

M Mohammed Dawoud Mota



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      </p>
    </div>
  </div>
 </noscript>
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/html>[ec2-user@ip-10-0-0-93 ~]$
```

# Introducing Today's Project!

## What is Amazon VPC?

Amazon VPC is a virtual network in AWS that lets us launch resources in a secure and isolated environment, giving control over subnets, IPs, routing, and access.

## How I used Amazon VPC in this project

I used Amazon VPC to host public and private subnets, launch EC2 instances, and test connectivity between them and to the internet securely.

## One thing I didn't expect in this project was...

I didn’t expect that the first ping between EC2 instances would fail, which showed me how important security group rules are for connectivity.

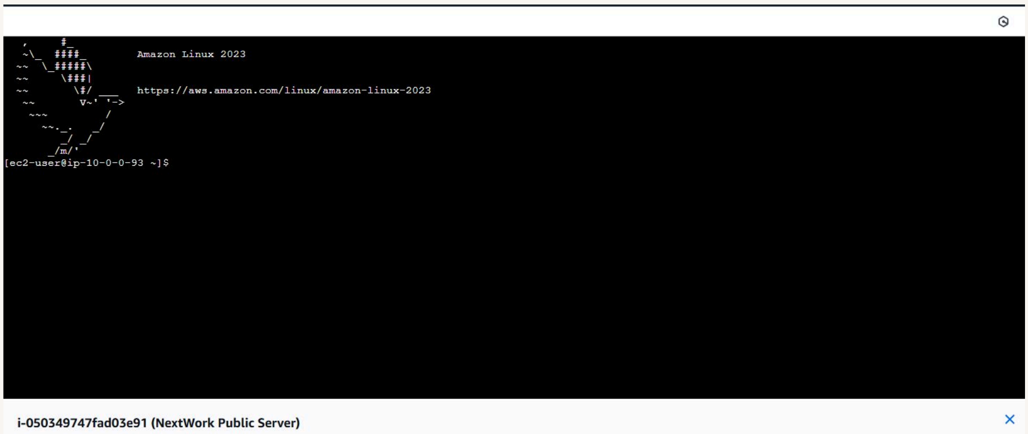
## This project took me...

This project took me 30 minutes.

# Connecting to an EC2 Instance

Connectivity means the ability of our resources, like EC2 instances and subnets, to communicate with each other and with the internet without any blocks or interruptions.

I was trying to connect to the public EC2 server in our VPC using the AWS Management Console.



# EC2 Instance Connect

EC2 Instance Connect is a service that lets us securely connect to our EC2 instances using a browser or SSH without needing to manage key pairs manually.

My first attempt at getting direct access to my public server resulted in an error because the attached security group did not have a rule to allow SSH access from anywhere.

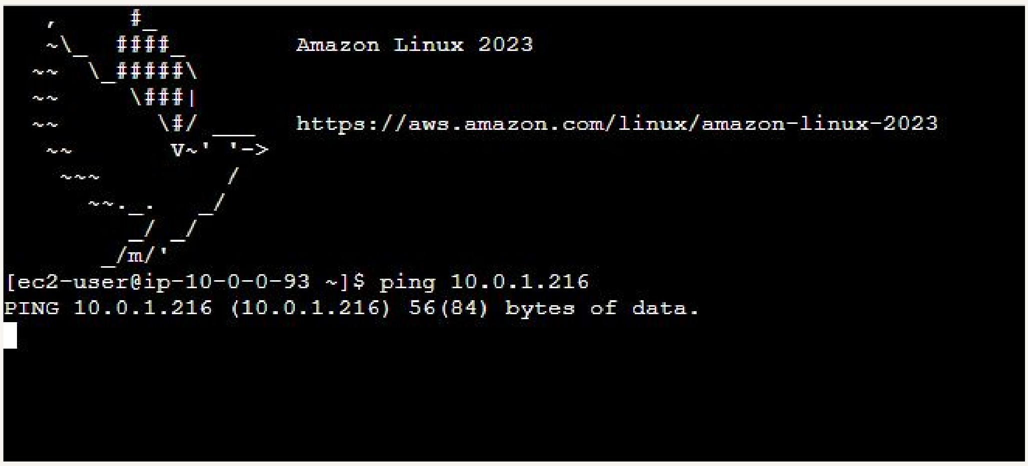
I fixed this error by going to the attached security group, clicking edit inbound settings, and adding a rule for SSH access from anywhere.

# Connectivity Between Servers

Ping is a tool that sends a small packet to a server to check if it responds. I used it to test if our EC2 instances could communicate with each other and confirm network connectivity.

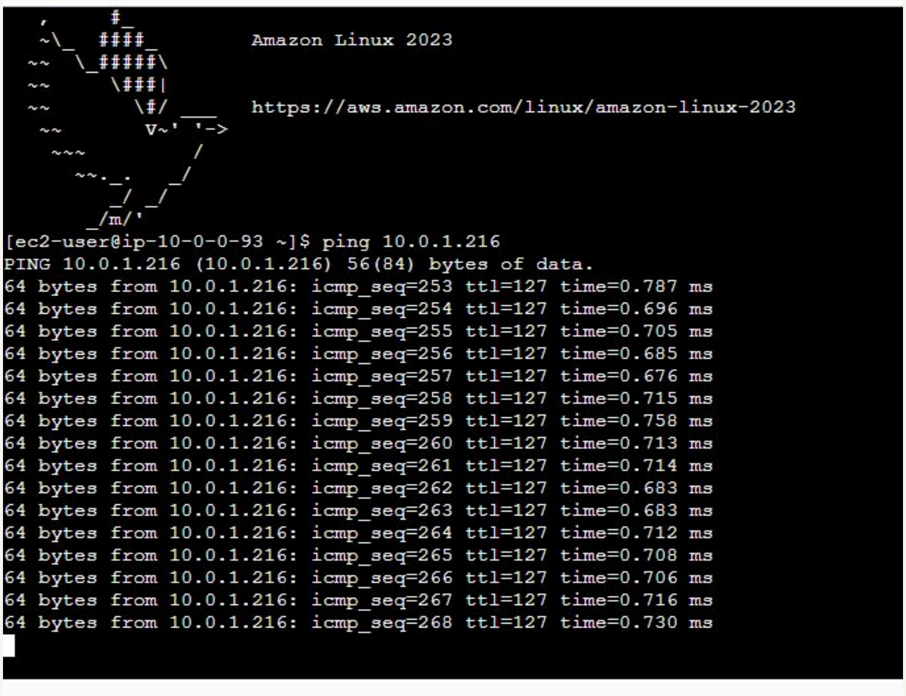
I ran the ping command followed by the private IP address of the other EC2 instance to test connectivity between them.

The first ping returned a timeout, which means the instances couldn’t communicate due to security rules or network configuration issues.



# Troubleshooting Connectivity

I resolved the error by updating the security group to allow ICMP traffic so the EC2 instances could successfully communicate with each other.



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   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-0-93 ~]$ ping 10.0.1.216
PING 10.0.1.216 (10.0.1.216) 56(84) bytes of data.
64 bytes from 10.0.1.216: icmp_seq=253 ttl=127 time=0.787 ms
64 bytes from 10.0.1.216: icmp_seq=254 ttl=127 time=0.696 ms
64 bytes from 10.0.1.216: icmp_seq=255 ttl=127 time=0.705 ms
64 bytes from 10.0.1.216: icmp_seq=256 ttl=127 time=0.685 ms
64 bytes from 10.0.1.216: icmp_seq=257 ttl=127 time=0.676 ms
64 bytes from 10.0.1.216: icmp_seq=258 ttl=127 time=0.715 ms
64 bytes from 10.0.1.216: icmp_seq=259 ttl=127 time=0.758 ms
64 bytes from 10.0.1.216: icmp_seq=260 ttl=127 time=0.713 ms
64 bytes from 10.0.1.216: icmp_seq=261 ttl=127 time=0.714 ms
64 bytes from 10.0.1.216: icmp_seq=262 ttl=127 time=0.683 ms
64 bytes from 10.0.1.216: icmp_seq=263 ttl=127 time=0.683 ms
64 bytes from 10.0.1.216: icmp_seq=264 ttl=127 time=0.712 ms
64 bytes from 10.0.1.216: icmp_seq=265 ttl=127 time=0.708 ms
64 bytes from 10.0.1.216: icmp_seq=266 ttl=127 time=0.706 ms
64 bytes from 10.0.1.216: icmp_seq=267 ttl=127 time=0.716 ms
64 bytes from 10.0.1.216: icmp_seq=268 ttl=127 time=0.730 ms
```

# Connectivity to the Internet

Curl is a command-line tool used to send requests to a server and see its response, which helps test connectivity and access to websites or APIs.

I used curl to test if our EC2 instance could reach the internet and receive a response from a web server, confirming outbound connectivity.

## Ping vs Curl

Ping checks if a server is reachable by sending packets, while curl requests data from a server and shows the response, testing both connectivity and service access.

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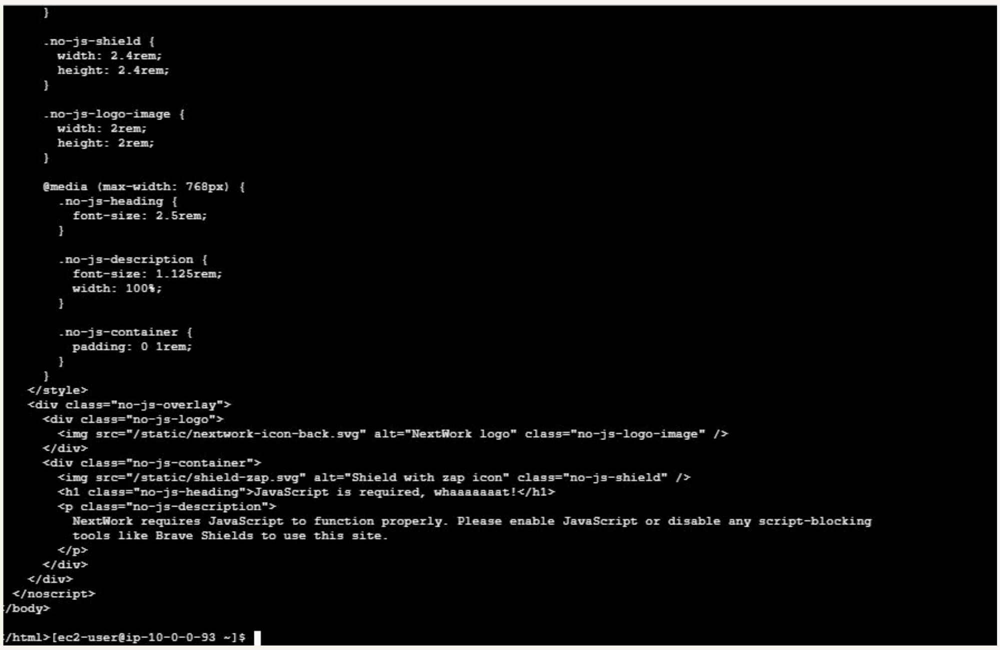
**Mohammed Dawoud M...**
NextWork Student



# Connectivity to the Internet

I ran curl followed by a public website URL to verify that our EC2 instance could access the internet and receive a response.



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